

Office with 25% solar behind the meter (real measured data)

Prepared for **Correctness test — real metered data** from 8,749 meter readings at 60-minute intervals, 99.9% of the period covered.

WHAT SITS BEHIND THIS METER

This site has **solar generation behind the meter** (1,400 kWp). Your grid meter therefore records only what you bought, not what the building used. Every figure in this report is the **building's actual load**, reconstructed from both series:

building load = grid meter + PV generation

Over the period the array generated **304,671 kWh**, of which 289,671 kWh was used on site and 15,000 kWh was exported. The building used **1,218,684 kWh** against the 914,013 kWh the meter recorded. Using the meter alone would have understated this site by 33%.

ELECTRICITY USED WHILE THE BUILDING WAS CLOSED

648,821 kWh = 155,717 EUR

That is **53% of everything this site bought last year**, used outside the hours you told us the building is in use. Not all of it is waste — some equipment has to run. The question this report answers is how much has to, and how you would find out.

LOAD WHILE THE BUILDING IS CLOSED

71 kW

Against 227 kW when people are in. 31% of the working-day load is still running with nobody there.

WHAT ONE KILOWATT IS WORTH

1,498 EUR/yr

Switch off 1 kW outside working hours and it saves this much every year. Ten kilowatts is ten times it.

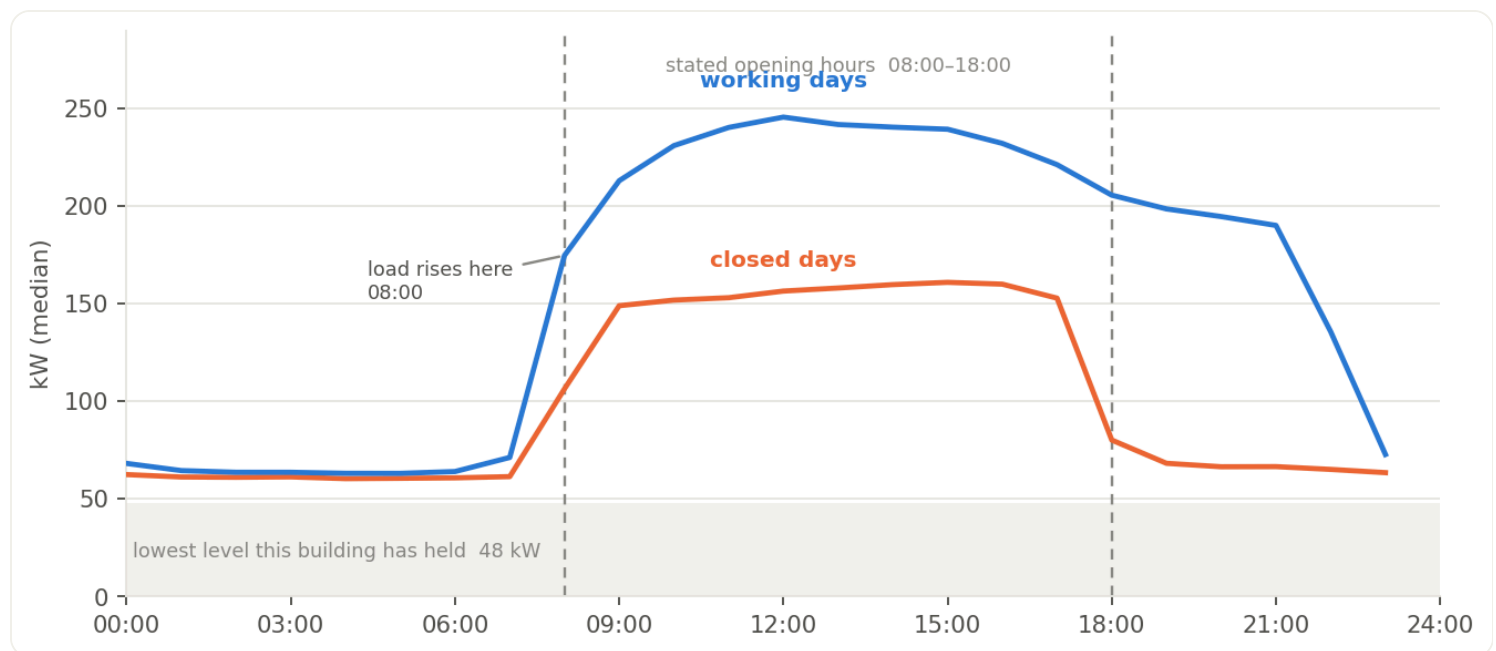
LOWEST THIS SITE HAS HELD

48 kW

Reached on 63 separate nights, so the equipment already installed can do it.

THE DAY, HOUR BY HOUR

Both lines sit at the same level all night. The building does not switch off — it turns up.



Median load at each time of day across the whole period, working days against closed days. The shaded band is the lowest level the site has actually held while closed — everything above it is load that was switched on.

WHERE TO LOOK FIRST

Five places, in the order a facilities team would check them.

1 The off step, and who is in the building after hours

Load stays up until 22:00, 4.0 hours past 18:00. Cleaning rounds and a late off step are the two usual causes, and one phone call to the cleaning contractor tells you which.

2 What is actually drawing the overnight load

Something holds 71 kW through the night. This report can say how much and when, not what. Server and comms rooms, lifts, external and car-park lighting, refrigeration and car-park ventilation are where it usually sits. A week with a clamp meter on the main sub-circuits settles it for good.

3 The days the building was supposed to be empty

24,853 kWh (5,965 EUR) was used across the 9 closed dates you listed. Those are the cheapest days to trial a change, because nothing that happens on them can inconvenience anybody.

These are places to check, not conclusions. This review reads the meter; it has not seen your plant. Each item names what the data shows and the usual cause, so whoever knows the building can confirm or rule it out in an afternoon.

WHAT A REDUCTION IS WORTH

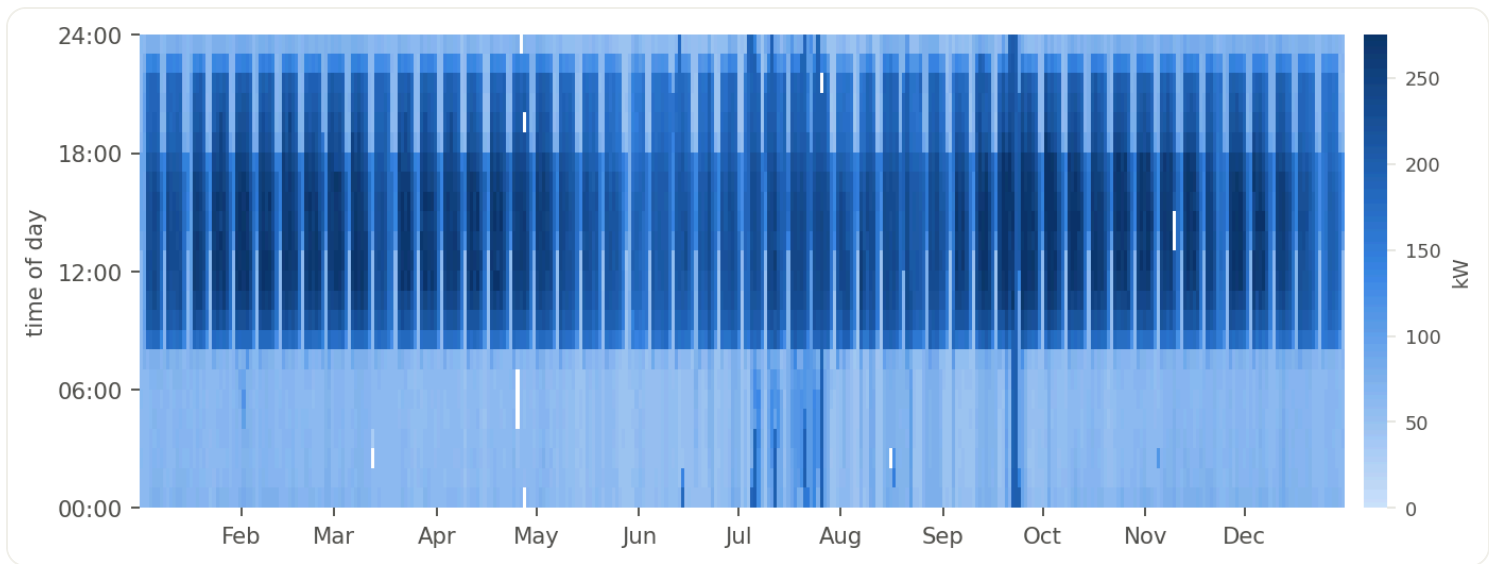
You pick the target. This converts it into money.

IF CLOSED-PERIOD LOAD FELL BY	NEW LEVEL	KWH SAVED A YEAR	EUR SAVED A YEAR
10% lower closed-period load	64.0 kW	44,379	10,651
25% lower closed-period load	53.3 kW	110,946	26,627
50% lower closed-period load	35.6 kW	221,893	53,254

Arithmetic on your own meter data: kW reduction × 6,241 closed hours × 0.240 EUR/kWh. This is not a prediction of what any particular measure would achieve — it converts a reduction you choose into the money it is worth.

EVERY HOUR OF THE PERIOD

One cell per hour. The dark band that never fades is the load that never goes away.



White means no reading was received. The pale vertical stripes are weekends. Read the picture bottom to top as one day, left to right as the year.

MEASURED Electricity used while the building was closed **648,821 kWh — 53.2% of the year's total**

Worth 155,717 EUR at the tariff supplied. The building was closed for 6,241 of 8,749 hours.

how: sum of readings outside the operating hours you supplied

DERIVED Load while closed **71.1 kW typical**

The building draws a median of 71.1 kW while closed, against 227.2 kW while occupied — 31% of the occupied load is still running when nobody is in the building.

how: median of all interval readings falling outside operating hours

DERIVED What each kilowatt is worth **1,498 EUR per year, per kW switched off**

Every 1 kW removed from the closed-period load saves $6,241 \text{ h} \times 0.240 \text{ EUR/kWh} = 1,498 \text{ EUR}$ a year. No target is assumed here — this is the conversion rate for whatever reduction you achieve.

how: closed hours x tariff

MEASURED Lowest level the building has held while closed **47.7 kW**

Reached on 63 separate closed periods, so the site is capable of it with the equipment already installed. Holding that level for the whole year would have saved 351,454 kWh (84,349 EUR).

how: 10th percentile of the per-night minimum load

DERIVED Actual vs stated operating hours **load rises 0.0 h early, falls 4.0 h late**

Stated hours are 08:00-18:00, but load crosses 10% of its daily range at 08:00 and stays there until 22:00.

how: median weekday load profile, 10% of daily range as the on/off threshold

DERIVED

Weekend load

94.3 kW average (42% of occupied load)

Weekend average against the occupied-hours average.

how: mean of Saturday+Sunday intervals / mean of occupied intervals

MEASURED

Consumption on days the building was stated empty

24,853 kWh over 9 days

Worth 5,965 EUR. These are the holiday and shutdown dates you supplied.

how: sum of readings on supplied holiday/shutdown dates

MEASURED

Peak demand

314.4 kW

Reached on 2017-10-24 at 14:00.

how: maximum interval reading converted to kW

HIGHEST OUT-OF-HOURS DAYS

DATE	DAY	KWH ABOVE THE SITE'S OWN LOWEST LEVEL	EUR
2017-09-23	Saturday	3,662	879
2017-02-20	Monday	2,867	688
2017-07-22	Saturday	2,509	602
2017-09-24	Sunday	2,300	552
2017-09-22	Friday	2,187	525
2017-07-04	Tuesday	2,148	515
2017-07-15	Saturday	2,118	508
2017-09-04	Monday	1,952	468
2017-11-24	Friday	1,926	462
2017-11-23	Thursday	1,794	431

CHECKS RUN BEFORE THIS REPORT WAS ISSUED

PASS	On-site generation and storage were asked about before the analysis ran	PV and battery both answered in the job file
PASS	The meter shows no generation the client did not declare	322 intervals are negative, totalling 15,000 kWh exported. Energy left the building, so something behind this meter generates.
PASS	PV generation is effectively zero while the sun is below the horizon	553.2 kWh (0.18% of generation) across 3,978 dark intervals
PASS	PV generation is never negative	0 negative interval(s)
PASS	Reconstructed building load is never negative	0 negative interval(s) after reconstruction
PASS	Building used more than the grid meter shows (self-consumption is positive)	1,218,684 kWh used vs 914,013 kWh bought
PASS	Self-consumed generation equals generation minus export	$304,671.0 - 15,000.3 = 289,670.7$ kWh
PASS	Grid meter total equals the sum of the rows used	meter 914,013.3097 kWh vs rows 914,013.3097 kWh
PASS	Building load equals meter + generation + discharge - charge	$914,013.3 + 304,671.0 + 0.0 - 0.0 = 1,218,684.3525$ vs 1,218,684.3525 kWh
PASS	Report total equals the reconstructed building load	report 1,218,684.3525 kWh vs reconstructed 1,218,684.3525 kWh
PASS	Rows used account for the raw file total (minus removed duplicates)	raw 914,013.31 kWh, used 914,013.31 kWh, 0 duplicate row(s) removed
PASS	Closed + occupied energy equals the total (no interval counted twice)	1,218,684.3525 vs 1,218,684.3525 kWh
PASS	Negative meter readings are accounted for as export	322 negative interval(s) = 15,000 kWh exported, added back as self-consumption before any figure below was computed
PASS	Interval coverage at least 95%	99.87% (11 intervals missing in 7 gap(s))
PASS	Achievable level is below the typical closed load	47.72 kW vs 71.11 kW
PASS	Avoidable energy is a strict subset of closed-period energy	351,453.9 of 648,821.0 kWh
PASS	Stated hours describe this building's actual pattern	closed-period load is 31% of occupied load
PASS	Avoidable energy below 40% of annual use (sanity ceiling)	28.8% of annual consumption

These run automatically every time the report is produced, and each one compares the report back to the file you sent. If any had failed you would not be reading this. The accompanying workbook lets you re-derive every headline number in Excel yourself, from your own readings, without taking anything here on trust.

DATA RECEIVED

Rows in file	8,749
Rows used	8,749
Reading interval	60 minutes
Coverage of the period	99.87%
Duplicate timestamps removed	0
Total energy in file	914,013.3 kWh
Timestamps	wall-clock local time
Tariff used	0.240 EUR/kWh (supplied by you)

GAPS IN THE DATA

FROM	TO	INTERVALS MISSING
2017-04-25 04:00	2017-04-25 07:00	3
2017-04-26 23:00	2017-04-27 01:00	2
2017-11-10 13:00	2017-11-10 15:00	2
2017-03-12 02:00	2017-03-12 03:00	1
2017-04-27 19:00	2017-04-27 20:00	1
2017-07-26 21:00	2017-07-26 22:00	1
2017-08-16 02:00	2017-08-16 03:00	1

Missing intervals are left missing. They are never estimated or filled in, because filling them would change your totals.

Prepared by Ali Mert Ozdemir. Figures marked MEASURED come straight from the file you supplied; DERIVED figures are arithmetic on it, with the formula shown beside each one. This review reports what the meter data shows. It is not an engineering survey and does not certify compliance with any standard.